Tiny Guests

01 Could a tiny guest be a helper?



Many bacteria help us. Some help make yogurt.

Look closer. A yogurt cup sits beside the oval guest in the book.

A little explanation. Bacteria are whole living cells. Many help us: some live in our gut and help with food, and others help make yogurt. Some can make people unwell. We can welcome the helpers and keep our hands clean.

Try this. Say one place where helpful bacteria can live.

02 What does a virus need to make copies?



A bacterium is a whole cell. A virus carries instructions in a coat.

Look closer. Compare the oval cell with the smaller round package and its coiled thread.

A little explanation. A virus is not a cell. It is a tiny package of instructions in a coat. It must borrow a living cell's machinery to make copies. Bacteria are much bigger than viruses, but both are too small to see with just your eyes.

Try this. Finish this thought: Instructions need something to carry them out, just as a virus needs...

03 How does soap help the water?



Soap loosens germs. Scrubbing and rinsing carry them away.

Look closer. Little rubs the lather over both mitten hands. The towel waits beside the sink.

A little explanation. Soap can pull apart the oily outer layers of many germs. Soap and scrubbing also lift germs off skin so running water can rinse them away. Plain water rinses off some, but soap helps remove more. Scrub for about 20 seconds, then rinse and dry.

Try this. Say the commands in order: Wet. Add soap. Scrub. Rinse. Dry.

04 What can your body remember?



Your body's defenders can learn and remember germs.

Look closer. Little keeps a guest's shape in the notebook.

A little explanation. Your immune system is your body's team of defenders. Some learn to recognize germs and remember their features next time. Antibiotics are different: these medicines work on bacteria but not viruses. Each kind of help has its own job.

Try this. Tell how learning a shape could help you recognize it again.